

پاسخ تمرین سری اول

2. Maximum likelihood of $f_Y(y; \lambda) = \lambda e^{-\lambda y} \quad y \geq 0$

likelihood function $L(\lambda; y_1, \dots, y_N) = \prod_{i=1}^N \lambda e^{-\lambda y_i}$

$$\Rightarrow \ln L(\lambda; y_1, \dots, y_N) = \sum_{i=1}^N (\ln \lambda - \lambda y_i) = N \ln \lambda - \lambda \sum_{i=1}^N y_i$$

$$\frac{\partial \ln L(\lambda; y_1, \dots, y_N)}{\partial \lambda} = \frac{N}{\lambda} - \sum_{i=1}^N y_i = 0 \Rightarrow \hat{\lambda} = \frac{N}{\sum_{i=1}^N y_i}$$

3. $f_Y(y; a) = a y^{a-1} \quad y \in (0,1) \quad a > 0$

$$L(a; y_1, \dots, y_N) = \prod_{i=1}^N a y_i^{a-1} \Rightarrow \ln L(a; y_1, \dots, y_N) = \sum_{i=1}^N \ln(a y_i^{a-1}) = \sum_{i=1}^N [\ln(a) + (a-1) \ln y_i]$$

$$= N \ln a + (a-1) \sum_{i=1}^N \ln y_i$$

$$\frac{\partial \ln L}{\partial a} = \frac{N}{a} + \sum_{i=1}^N \ln y_i = 0 \Rightarrow \hat{a}_{ML} = -\frac{N}{\sum_{i=1}^N \ln y_i} = -\frac{1}{\frac{1}{N} \sum_{i=1}^N \ln y_i}$$

4. $y_n \sim N(\mu, 1) \quad \mu \sim N(0, \sigma_\mu^2)$

Since both the prior and likelihood are Normal, the posterior will also be Normal.

$$p(y_1, y_2, \dots, y_N / \mu) \propto \prod_{n=1}^N N(y_n / \mu, 1)$$

$$N(y_n / \mu, 1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(y_n - \mu)^2}{2}\right) \Rightarrow p(y_1, \dots, y_N) \propto \exp\left(-\frac{1}{2} \sum_{i=1}^N (y_i - \mu)^2\right)$$

$$\Rightarrow p(y_1, \dots, y_N) \propto \exp\left(-\frac{1}{2} \left(\sum_{n=1}^N y_n^2 - 2\mu \sum_{n=1}^N y_n + N\mu^2\right)\right)$$

$$p(\mu) = N(\mu / 0, \sigma_\mu^2) = \frac{1}{\sqrt{2\pi\sigma_\mu^2}} \exp\left(-\frac{\mu^2}{2\sigma_\mu^2}\right)$$

Bayes' Theorem:

$$p(\mu / y_1, \dots, y_N) \propto p(y_1, y_2, \dots, y_N / \mu) p(\mu)$$

$$p(\mu / y_1, y_2, \dots, y_N) \propto \exp\left(-\frac{1}{2} \left(\sum_{n=1}^N y_n^2 - 2\mu \sum_{n=1}^N y_n + N\mu^2\right)\right) \exp\left(-\frac{\mu^2}{2\sigma_\mu^2}\right)$$

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terms involve μ after combination:

$$-\frac{1}{2} \left(N + \frac{1}{\omega_{\mu}^2} \right) \mu^2 + \mu \left(\sum_{n=1}^N y_n \right)$$

Completing square:

$$P(\mu | y_1, y_2, \dots, y_N) \propto \exp \left(-\frac{1}{2} \left(N + \frac{1}{\omega_{\mu}^2} \right) \left(\mu - \frac{\sum_{n=1}^N y_n}{N + \frac{1}{\omega_{\mu}^2}} \right)^2 \right)$$

$$\Rightarrow \left. \begin{aligned} \omega_{\text{posterior}}^2 &= \frac{1}{N + \frac{1}{\omega_{\mu}^2}} \\ \mu_{\text{posterior}} &= \frac{\sum_{n=1}^N y_n}{N + \frac{1}{\omega_{\mu}^2}} = \frac{N\bar{y}}{N + \frac{1}{\omega_{\mu}^2}} \end{aligned} \right\} P(\mu | y_1, y_2, \dots, y_N) \sim N \left(\frac{N\bar{y}}{N + \frac{1}{\omega_{\mu}^2}}, \frac{1}{N + \frac{1}{\omega_{\mu}^2}} \right)$$

b) The Map estimator is the value of μ that maximizes the posterior distribution. posterior is Gaussian so the Map estimator is mode of Gaussian which is same as mean:

$$\hat{\mu}_{\text{MAP}} = \frac{N\bar{y}}{N + \frac{1}{\omega_{\mu}^2}}$$

5) Assume that δ is a known constant.

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with the probability density function:

$$f(x; \beta, \delta) = \beta \delta x^{\delta-1} e^{-\beta x^{\delta}}, x > 0, 0 \text{ otherwise.}$$

1. Derive the maximum likelihood estimator (MLE) of β , denoted as $\hat{\beta}$.

Maximum Likelihood Estimator (MLE) of β for the given distribution

The probability density function (PDF) is given as:

$$f(x; \beta, \delta) = \beta \delta x^{\delta-1} e^{-\beta x^{\delta}}, x > 0$$

Let the observed values be $X_1, X_2, \dots, X_n$. The likelihood function $L(\beta, \delta)$ is the joint probability of observing the sample:

$$L(\beta, \delta) = \prod_{i=1}^n f(X_i; \beta, \delta) = \prod_{i=1}^n \beta \delta X_i^{\delta-1} e^{-\beta X_i^{\delta}}$$

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Now, to derive the MLE of β , we will first take the natural logarithm of the likelihood function to simplify the calculations. This is called the log-likelihood function:

$$\log L(\beta, \delta) = \sum_{i=1}^n (\log(\beta) + \log(\delta) + (\delta - 1)\log(X_i) - \beta X_i^\delta)$$

Simplifying further:

$$\log L(\beta, \delta) = n\log(\beta) + n\log(\delta) + (\delta - 1) \sum_{i=1}^n \log(X_i) - \beta \sum_{i=1}^n X_i^\delta$$

To find the MLE of β , we differentiate the log-likelihood function with respect to β and set the derivative equal to zero:

$$\frac{\partial}{\partial \beta} \log L(\beta, \delta) = \frac{n}{\beta} - \sum_{i=1}^n X_i^\delta = 0$$

Now solving for β :

$$\begin{aligned} \frac{n}{\beta} &= \sum_{i=1}^n X_i^\delta \\ \beta &= \frac{n}{\sum_{i=1}^n X_i^\delta} \end{aligned}$$

Thus, the MLE of β is:

$$\hat{\beta} = \frac{n}{\sum_{i=1}^n X_i^\delta}$$

6) Let $\tau > 0$, and let $X_1, X_2, \dots, X_n$ be a random sample drawn from a distribution with the probability density function:

$$f(x; \tau) = \tau e^{-\tau(x-3)}, x > 3, 0 \text{ otherwise}$$

1. Derive the maximum likelihood estimator (MLE) of τ , denoted as $\hat{\tau}$.
2. Given a sample of size $n = 4$ with observed values $X_1 = 3.15, X_2 = 3.65, X_3 = 3.10$, and $X_4 = 3.30$, compute the maximum likelihood estimate of τ .

Maximum Likelihood Estimator (MLE) of τ for the given distribution

The second problem involves the following PDF:

$$f(x; \tau) = \tau e^{-\tau(x-3)}, x > 3$$

The likelihood function $L(\tau)$ for a sample $X_1, X_2, \dots, X_n$ is:

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$$L(\tau) = \prod_{i=1}^n \tau e^{-\tau(X_i-3)}$$

Taking the natural logarithm to form the log-likelihood:

$$\log L(\tau) = \sum_{i=1}^n (\log(\tau) - \tau(X_i - 3))$$

Simplifying:

$$\log L(\tau) = n \log(\tau) - \tau \sum_{i=1}^n (X_i - 3)$$

To find the MLE of τ , we differentiate the log-likelihood with respect to τ and set the derivative equal to zero:

$$\frac{\partial}{\partial \tau} \log L(\tau) = \frac{n}{\tau} - \sum_{i=1}^n (X_i - 3) = 0$$

Solving for τ :

$$\begin{aligned} \frac{n}{\tau} &= \sum_{i=1}^n (X_i - 3) \\ \tau &= \frac{n}{\sum_{i=1}^n (X_i - 3)} \end{aligned}$$

Thus, the MLE of τ is:

$$\hat{\tau} = \frac{n}{\sum_{i=1}^n (X_i - 3)}$$

3. Compute the MLE of τ for the given sample

Given the sample:

$$X_1 = 3.15, X_2 = 3.65, X_3 = 3.10, X_4 = 3.30$$

The number of observations $n = 4$, and we need to compute the sum:

$$\sum_{i=1}^n (X_i - 3) = (3.15 - 3) + (3.65 - 3) + (3.10 - 3) + (3.30 - 3)$$

Let's calculate the sum and then use the formula for $\hat{\tau}$:

$$\hat{\tau} = \frac{4}{\sum_{i=1}^4 (X_i - 3)}$$

Res = 3.33